

# AURORA-12

## A fully synthetic benchmark for evidence-linked poster generation

PostEx Synthetic Research Collective

Fictional Methods Laboratory · Example Research Campus · no real authors or institutions

POSTEX-SYNTH-2026-001 · fully synthetic · CC0-1.0 · no DOI

Fictional lab logo

1,240  
synthetic  
profiles

12  
anonymous  
model features

4  
deterministic  
cohort  
simulations

### Communication test

Can a realistic multi-cohort manuscript exercise evidence-linked poster generation without redistributing a real paper or implying biomedical evidence?

Evidence: Abstract and Introduction, pp. 1-2

### Deterministic workflow

- Fixed seed generated 1,240 profiles
- 480 anonymous transcript-like features
- Nested elastic-net selection
- 12-feature score and shift testing

#### FOUR SYNTHETIC COHORTS

- Four deterministic synthetic cohorts
- Discovery n=620 · Internal validation n=310
  - External A n=186 · External B n=124
  - No participants, specimens or external datasets

Evidence: Abstract and Methods, pp. 1-2

### Generation to validation

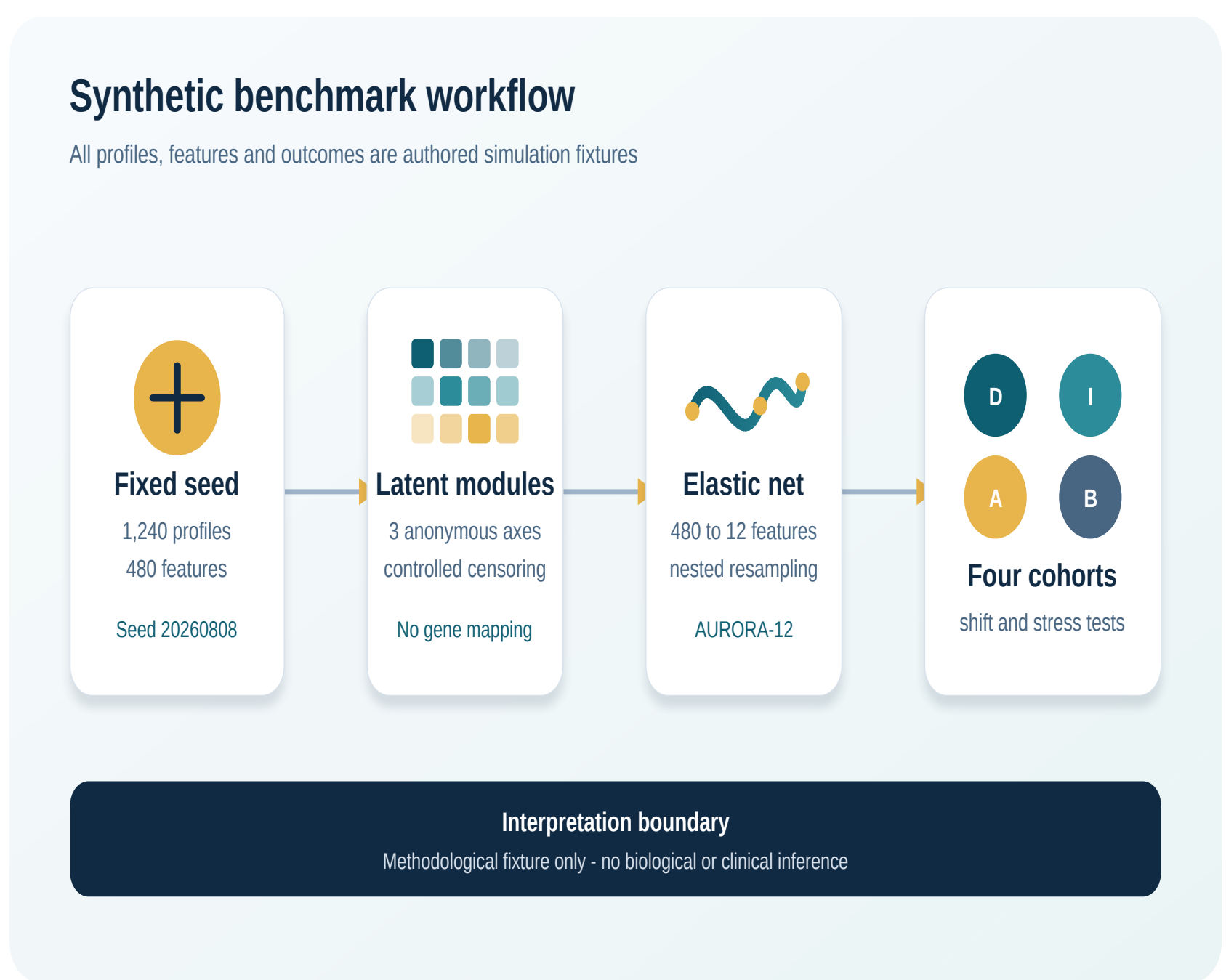


Figure 1 · Deterministic generation and validation workflow  
Authored synthetic fixture · manuscript p. 2

### External C-indices remained 0.75 and 0.74

Synthetic baseline: 0.66 and 0.65 · software fixture, not clinical validation

Evidence: Results, pp. 1 and 3; Figure 2

### Discrimination across shifts



Figure 2 · Authored discrimination values across cohort shifts  
Authored synthetic fixture · manuscript p. 3

### Calibration and stress boundary

- External calibration slopes were 0.97 and 0.95.
- External absolute errors were 0.031 and 0.038.
- Aggregate coefficient-sign stability was 92% across 500 resamples.
- The most severe pre-specified shift reduced C-index by 0.04.

Evidence: Results, pp. 1, 3-4; Figures 3-4

External validation retained discrimination: C-index 0.75 / 0.74

### Calibration & stability

92%

coefficient-sign stability

0.97 / 0.95

external calibration slopes

0.031 / 0.038

external absolute errors

Evidence: Stability and calibration, p. 4

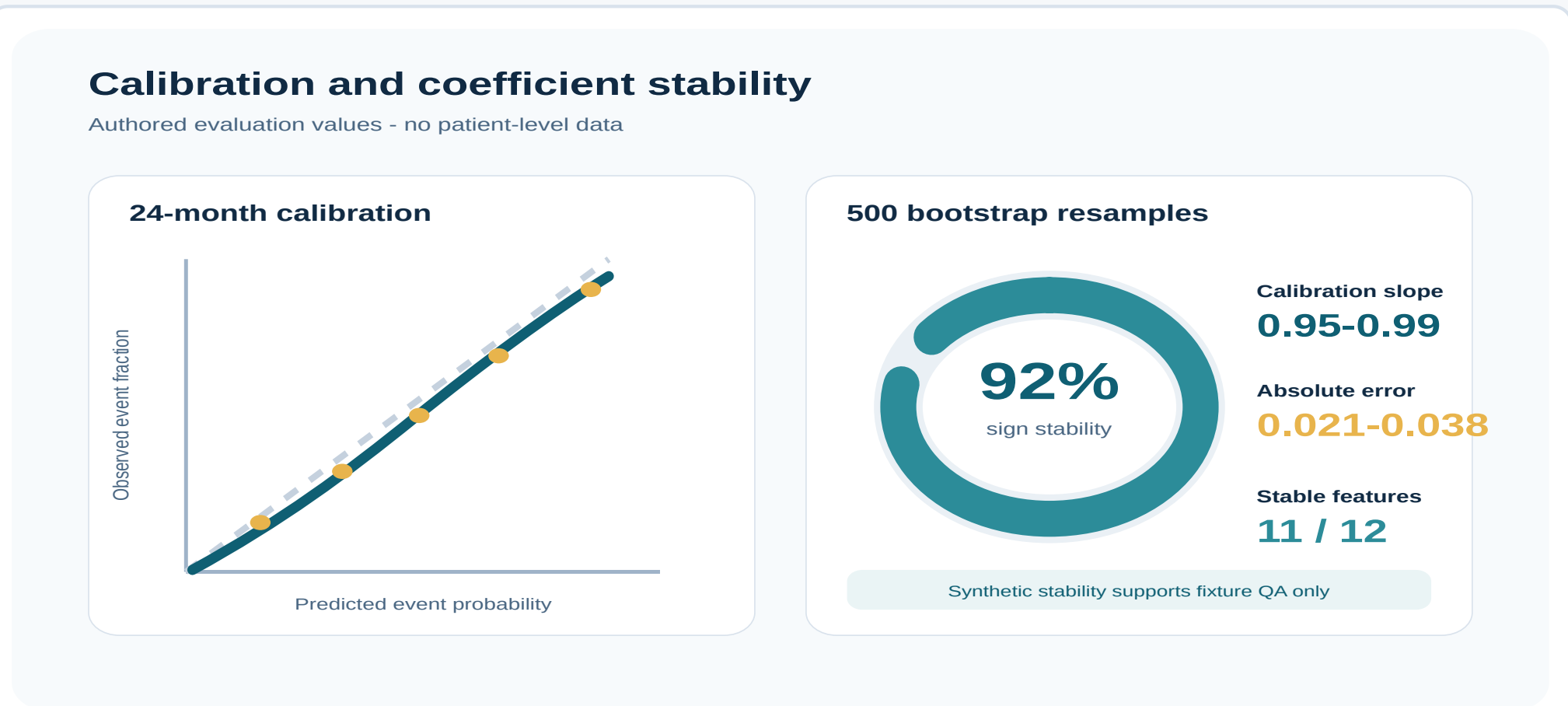


Figure 3 · Calibration and coefficient-sign stability  
Authored synthetic fixture · manuscript p. 4

### Pre-specified stress test

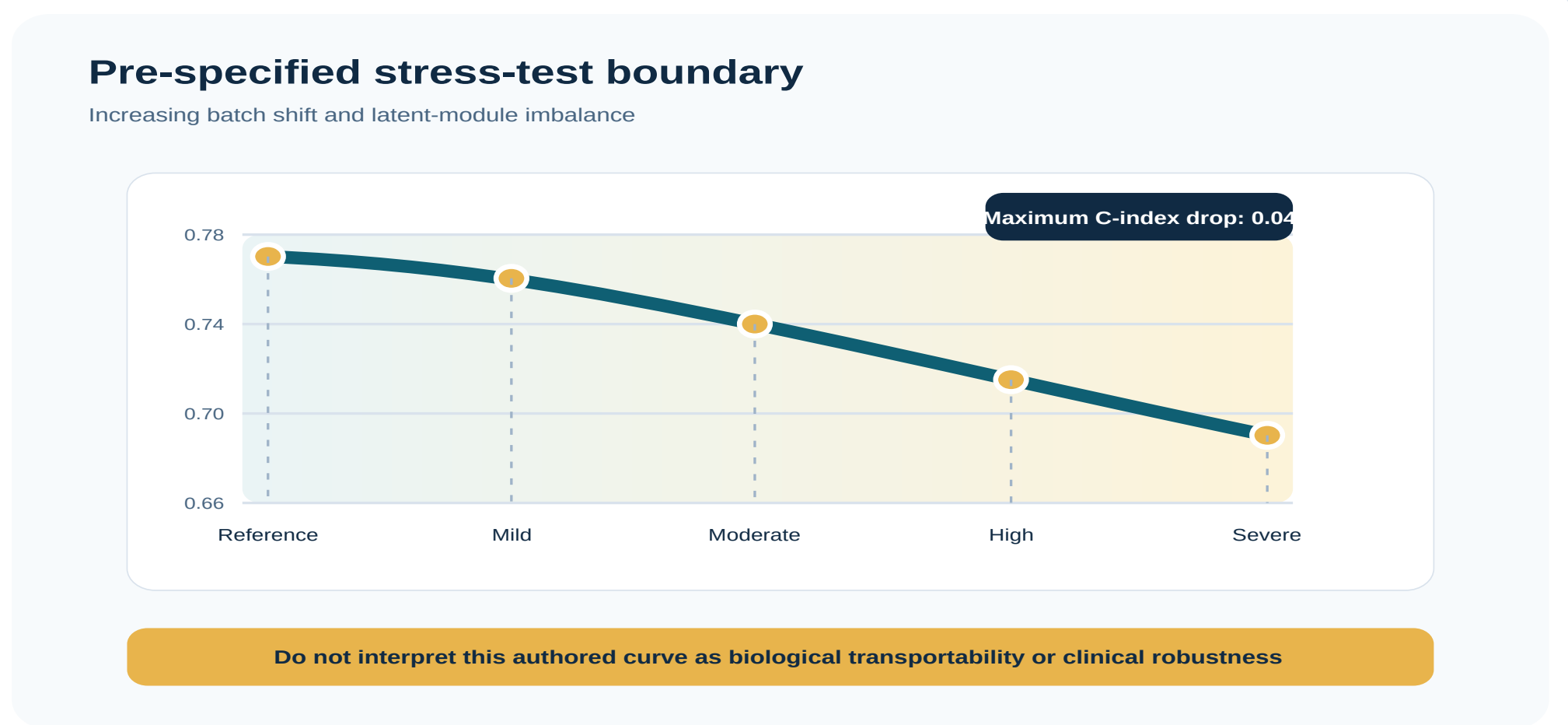


Figure 4 · Stress-test boundary under increasing shift  
Authored synthetic fixture · manuscript p. 4

### Supported conclusion

#### A rights-safe methodological test fixture

- No biological, prognostic or clinical inference
- All cohorts, values and identities are fictional

Evidence: Discussion and Conclusion, p. 5